Chapter 3 — Data and Methodology

3.1 Research Design

The empirical strategy of this dissertation is designed to test the four hypotheses introduced in Chapter 1 at the scales at which they make claims. H1 and H2 concern the shape of the education–fertility gradient at the population level and are tested on a panel of European countries. H3 concerns cross-country variation in that gradient and the relative weight of cultural versus economic explanations, tested using country-level covariates. H4 concerns the individual-level mechanism by which education depresses fertility, tested on longitudinal data from a single British birth cohort. The design therefore moves from macro to micro, with each scale of analysis matched to a method appropriate to its data structure and inferential claim.

Three analytical datasets are constructed for the four hypotheses. Dataset 1 is an unbalanced country–year–education panel of period total fertility rates for 21 European countries between 2007 and 2024, used to test H1 and H2. Dataset 2 is a country-level cross-section of 21 observations combining derived gradient outcomes from Dataset 1 with cultural, economic, and regional predictors from multiple sources, used to test H3. Dataset 3 is the individual-level analytical sample drawn from the 1970 British Cohort Study, used to test H4. Table 3.1 summarises the data sources.

Table 3.1. Data sources

Dataset	Source	Unit of observation	Coverage	Role
demo_faeduc	Eurostat	Country × year × age × ISCED	21 countries, 2007–2024	Births numerator (H1, H2)
edat_lfse_03	Eurostat	Country × year × age × ISCED × sex	21 countries, 2007–2024	Education shares of population (H1, H2)

demo_pjangroup	Eurostat	Country × year × age × sex	21 countries, 2007–2024	Female population denominator (H1, H2)
V-Dem v16	V-Dem Institute	Country × year	21 countries, 2007–2024	Institutional cultural predictors (H3)
GDP per capita	OECD	Country × year	18 of 21 countries	Economic predictor (H3)
Female LFP rate	OECD	Country × year	19 of 21 countries	Economic predictor (H3)
SP.DYN.CONU.ZS	World Bank	Country × year	12 of 21 countries	Economic predictor (H3)
ESS rounds 1–11	European Social Survey	Individual, pooled across rounds	20 of 21 countries	Attitudinal cultural predictors (H3)
BCS70	UK Data Service (SN 2666, 3723, 3535, 6095, 5558, 7473, 9347)	Individual, longitudinal	≈16,500 born April 1970, sweeps at birth, 10, 16, 30, 42, 51	Individual-level mediation (H4)

Coverage figures refer to the analytical window after cleaning and merging. ESS coverage applies to the country-level aggregate composites. BCS70 Study Numbers refer to UK Data Service release identifiers.

The analytical strategy diverges from the original research proposal in three important respects, each of which is documented transparently here. First, the primary fertility outcome is period education-stratified TFR (eTFR) computed from Eurostat rather than completed cohort fertility from the Wittgenstein Centre Data Explorer, because the Wittgenstein R interface exposes only projection data (2020–2100) and not the historical reconstruction series that the proposal assumed would be accessible in machine-readable form. Second, the cross-country sample is 21 rather than the 35 projected, because of country-specific gaps in Eurostat's education-stratified fertility reporting. Third, H3 originally proposed using the European Social Survey (ESS) for country-level attitudinal aggregates. During analysis it became apparent that the five V-Dem institutional indices deployed as cultural predictors measured institutional quality rather than individual attitudes, creating a construct validity gap between the cultural transmission theory (which predicts that individual beliefs and norms drive fertility behaviour) and the indicators used to test it. To address this, ESS data were incorporated as an analytical refinement: three attitudinal composite predictors derived from eight ESS core-module items were added to Dataset 2 alongside the V-Dem institutional predictors. This distinction between V-Dem institutional and ESS attitudinal cultural indicators is central to the H3 analysis and is discussed in Section 3.7.

3.2 Construction of Dataset 1: Education-Stratified Period Fertility

3.2.1 Data sources and the Wittgenstein–Eurostat pivot

Three Eurostat series form the empirical backbone of Dataset 1. The demo_faeduc series provides counts of live births by mother's five-year age group and ISCED 2011 educational attainment, by country and year. The edat_lfse_03 series provides the educational composition of the female population (shares by ISCED level) within broader age bands, by country and year. The demo_pjangroup series provides total female population counts by five-year age group, by country and year. The education-specific female population in each age cell is estimated by multiplying the total population (from demo_pjangroup) by the education share (from edat_lfse_03), under the standard demographic assumption that educational composition is approximately uniform within five-year subgroups of the 35–44 and 45–54 ten-year reporting bands used in edat_lfse_03 for older ages.

This assumption is defensible because educational attainment in European populations stabilises well before age 35; the distributions at 35–39 and 40–44 are sufficiently similar within the reporting bands that the proportional split introduces only minor approximation error. The edat_lfse_03 series for the 45–54 band is applied to the demo_pjangroup 45–49 cell only, so the 50–54 portion of the population, which lies outside the reproductive age range, does not enter the denominator.

3.2.2 Country sample

The country sample was determined by the intersection of demo_faeduc and edat_lfse_03 after exclusion of aggregate units (European Union aggregates, Euro area aggregates, and so on) and after audit for the presence of education-disaggregated fertility data. Table 3.2 documents the four-stage walk-down from the initial 35-country intersection to the final 21-country analytical sample.

Table 3.2. Country sample evolution

Stage	N	Reason	Log entry
Initial	35	All European countries in intersection of demo_faeduc and edat_lfse_03	Entry 11
Stage 1	23	12 countries report fertility only as "All ISCED 2011 levels" with no education breakdown (Bulgaria, Switzerland, Cyprus, Germany, France, Iceland, Italy, Lithuania, Luxembourg, Malta, the Netherlands, Ireland)	Entries 13–14
Stage 2	22	Bosnia and Herzegovina dropped: no age-disaggregated population data in demo_pjangroup	Entry 17
Final	21	Montenegro dropped: demo_faeduc covers 2007–2010, edat_lfse_03 covers 2011–2020 — zero year overlap makes ASFR construction infeasible	Entry 21
Note	—	The United Kingdom is excluded from H1–H3 due to post-Brexit gaps in demo_faeduc reporting, but is represented in the dissertation through the BCS70 cohort in H4	Entry 12

All exclusions reflect data-availability constraints rather than analytical choices. The final 21-country sample spans Nordic (Denmark, Finland, Norway, Sweden), Eastern European (Croatia, Czechia, Estonia, Hungary, Latvia, Poland, Romania, Serbia, Slovakia, Slovenia), Mediterranean (Greece, Portugal, Spain), and other (Austria, Belgium, North Macedonia, Türkiye) fertility regimes.

The final panel is unbalanced. Austria contributes six years (2007–2012), Belgium seven (2007–2014), and Latvia eight (2017–2024). Spain and Türkiye each contribute twelve years. Sixteen of 21 countries have between 13 and 18 years of usable data. The two-way fixed-effects estimator used for H1 and H2 accommodates unbalanced panels naturally; a balanced-subsample robustness check restricted to the eight countries with full eighteen-year coverage is reported in Section 3.9.

3.2.3 Education harmonisation

Educational attainment is harmonised to three ISCED 2011 categories: low (ISCED 0–2, less than primary through lower secondary), medium (ISCED 3–4, upper secondary and post-secondary non-tertiary), and high (ISCED 5–8, tertiary). This three-category scheme is the most granular grouping for which education-stratified fertility data are jointly available across all 21 countries in the sample. Finer ISCED-level categorisation is reported by a subset of countries but cannot be harmonised across the full sample without imputation. The three-category scheme is also the standard convention in European education-fertility research and produces interpretable, theoretically meaningful tiers (Skirbekk, 2008).

3.2.4 ASFR and eTFR construction

Education-specific age-specific fertility rates (ASFR) are constructed as the ratio of births to the estimated female population within each country × year × five-year age group × education cell. The eTFR for each country × year × education cell is then computed as:

$$\mathrm{eTFR}_r = 5 \times \Sigma_a \, \mathrm{ASFR}_{r,a}$$

where r indexes the education tier and a the seven five-year age groups between 15–19 and 45–49. The factor of five converts the single-year-equivalent rate to a synthetic-cohort equivalent, so that eTFR is interpretable on the same scale as the standard total fertility rate. The resulting Dataset 1 comprises 933 country–year–education observations. Table 3.3 summarises the variables.

Table 3.3. Variables in Dataset 1 (etfr_data.rds)

Variable	**Definition**	**Range / coding**
country	European country (ISO short name)	21 levels

year	Calendar year of observation	2007–2024 (unbalanced)
education	Harmonised ISCED 2011 attainment category	low (0–2), medium (3–4), high (5–8)
n_age_groups	Number of five-year age groups contributing to the eTFR sum for the cell; used to flag incomplete cells	6 or 7 (7 = complete)
etfr	Education-stratified period total fertility rate = 5 × Σ ASFR across seven age groups 15–49	0.36–3.39 (children per woman)
log_etfr	Natural log of eTFR; outcome variable in H1 and H2 specifications	approx. −1.0 to 1.2

N = 933 country × year × education observations across 21 countries, 2007–2024. log_etfr is the outcome variable in all H1 and H2 specifications. n_age_groups = 6 in a small number of early Austria cells where the 45–49 group is absent; all other cells are complete at n_age_groups = 7.

3.2.5 Gradient shape classification

Descriptive analysis (Chapter 4) revealed that the education–fertility gradient is non-monotonic in 12 of the 21 countries, with medium-educated women having lower fertility than both low- and high-educated women. A further two countries show an inverted-bottom pattern (low-educated women having fertility equal to or lower than medium-educated women). To support both the descriptive presentation and the heterogeneity-robust robustness checks, each country is classified into one of four populated gradient-shape categories on the basis of mean eTFR by education tier across 2007–2024:

Monotonic negative (low > medium > high): five countries (Austria, Portugal, Spain, Sweden, Türkiye). Inverted-bottom (low ≤ medium): two countries (Belgium, Denmark). Composition-driven U-shape (j_curve_composition: high > medium AND share of women aged 25–39 in the low-education tier < 12%): eight countries (Croatia, Czechia, Estonia, Finland, Latvia, Poland, Slovakia, Slovenia). Broad U-shape (j_curve_broad: high > medium AND share_low ≥ 12%): six countries (Greece, Hungary, North Macedonia, Norway, Romania, Serbia). A fifth residual category (other_flat) exists in the classification scheme but contains no countries in this sample.

The 12% threshold on the low-education share was chosen empirically as the natural break in the bimodal distribution of share_low across the 21 countries, separating countries where the small low-education tier plausibly reflects the over-representation of minority high-fertility populations (composition-driven U-shape) from those where the low tier is large enough that the elevated fertility reflects a structural feature of the gradient. The substantive findings for H1 and H2 are robust to variation in this threshold, as demonstrated in the robustness checks in Section 3.9.

3.3 Construction of Dataset 2: Country-Level Outcomes and Predictors

3.3.1 Derived outcomes

Dataset 2 takes the 21 countries as units of observation. The primary outcome variable, gradient_steepness, is defined as:

$$\textit{gradient\_steepness} = \text{mean}(\text{eTFR\_}\textit{low}) - \text{mean}(\text{eTFR\_}\textit{high})$$

computed across all available country–year cells in 2007–2024. Positive values indicate the conventional negative gradient (low > high); values near zero or negative indicate flat or inverted gradients. A secondary binary outcome, u_shape, indicates whether mean eTFR_medium is lower than both mean eTFR_low and mean eTFR_high. The categorical shape variable from Section 3.2.5 is also retained for descriptive purposes.

3.3.2 Institutional and attitudinal cultural predictors

Cultural and institutional predictors are drawn from two sources that operationalise distinct dimensions of culture, a distinction that emerged as analytically important during the study. The Varieties of Democracy (V-Dem) v16 dataset provides five institutional indices: liberal democracy (v2x_libdem), women's political empowerment (v2x_gender), freedom of religion (v2clrelig), electoral democracy (v2x_polyarchy), and egalitarian democracy (v2x_egaldem). Each index is computed as the country-level mean across all available years in 2007–2024.

These indices capture the formal institutional structure of a country's political and social environment rather than the beliefs and attitudes of individual citizens.

The five indices are highly collinear, correlations among libdem, polyarchy, and egaldem exceed $r = 0.98$, reflecting the fact that they capture overlapping dimensions of democratic institutional quality. Each Bayesian model in Stage 2 therefore includes at most two V-Dem predictors, following the standard practice of avoiding near-perfectly collinear regressors in the same specification.

The European Social Survey (ESS), rounds 1–11 (2002–2023), provides individual-level attitudinal data that were aggregated to country-level means to construct three composite cultural predictors. These composites operationalise the individual attitude dimensions that the cultural transmission theory of fertility decline (Bongaarts and Watkins, 1996; Colleran, 2016) identifies as the proximate channels through which low-fertility norms spread through populations. The secularisation composite (ess_secular) combines reversed religiosity (rlgdgr, 0–10 scale) and religious attendance (rlgatnd, rescaled to 0–10), with higher values indicating less religious belief and practice. The gender egalitarianism composite (ess_gender_egal) combines two reversed gender-role attitude items: agreement that women should cut down paid work for family (wmcpwrk) and that men should have priority for scarce jobs (mnrgtjb), both from 1–5 scales, with higher values indicating more egalitarian attitudes. The autonomy-versus-conformity composite (ess_autonomy) is computed from four Schwartz human values items as (ipcrtiv + impfree) − (imptrad + ipfrule), where positive values indicate net individualism and negative values net conformism. All ESS items are drawn from the core module, which is fielded in every round, providing maximum temporal and country coverage. ESS data were read from a locally stored integrated file and filtered to the 20 target countries (North Macedonia is not in ESS) before aggregation.

The combination of V-Dem institutional and ESS attitudinal predictors enables the H3 analysis to distinguish between two versions of the cultural hypothesis: one in which culture operates through formal institutions (the V-Dem channel) and one in which it operates through individual attitudes and social learning (the ESS channel). To my knowledge, this distinction has not been tested empirically in the education–fertility gradient literature.

3.3.3 Economic predictors and country harmonisation

Economic predictors include GDP per capita in constant US dollars from the OECD (country-level mean across 2007–2024, available for 18 of 21 countries; Serbia, North Macedonia, and Türkiye are not OECD members), female labour-force participation rate from the OECD (19 of 21 countries), and modern contraceptive prevalence (indicator SP.DYN.CONU.ZS from the World Bank Development Indicators, downloaded as a local CSV and averaged across available years, available for 12 of 21 countries). A regional indicator (post_socialist) takes the value one for countries with formerly socialist Eastern European political histories.

Country names are harmonised across all sources using the countrycode R package. Missingness across predictors is handled differently in the two analytical stages: the XGBoost model (Stage 1) uses all 21 countries and handles missing values natively; the Bayesian models (Stage 2) use list-wise complete-case deletion, yielding an effective sample of 18 countries after dropping North Macedonia (missing all ESS and GDP), Serbia (missing GDP and ess_gender_egal), and Türkiye (missing GDP). Table 3.4 lists all Dataset 2 variables.

Table 3.4. Variables in Dataset 2 (dataset2_full.rds)

Variable	Definition	Source / type
gradient_steepness	Mean eTFR_low − mean eTFR_high across 2007–2024	Derived (primary outcome)

u_shape	Indicator: 1 if mean eTFR_medium < both eTFR_low and eTFR_high	Derived (binary)
shape	Categorical gradient classification: monotonic_negative, inverted_bottom, j_curve_composition, j_curve_broad, other_flat	Derived (categorical)
gender_equal	V-Dem women political empowerment index (v2x_gender); 2007–2024 mean	V-Dem institutional
relig_free	V-Dem freedom of religion index (v2clrelig); 2007–2024 mean	V-Dem institutional
libdem	V-Dem liberal democracy index (v2x_libdem); 2007–2024 mean	V-Dem institutional
polyarchy	V-Dem electoral democracy index (v2x_polyarchy); 2007–2024 mean	V-Dem institutional
egaldem	V-Dem egalitarian democracy index (v2x_egaldem); 2007–2024 mean	V-Dem institutional
ess_secular	ESS composite: mean of reversed religiosity (rlgdgr) and rescaled attendance (rlgatnd); higher = more secular	ESS attitudinal
ess_gender_egal	ESS composite: mean of reversed wmcpwrk and mnrgtjb; higher = more egalitarian gender attitudes	ESS attitudinal (1 miss.)
ess_autonomy	ESS composite: (ipcrtiv + impfree) − (imptrad + ipfrule); higher = more individualist	ESS attitudinal
gdp_percap	GDP per capita, constant USD; 2007–2024 mean (OECD)	Economic (3 miss.)
flfp	Female labour-force participation rate; 2007–2024 mean (OECD)	Economic (2 miss.)
contraceptive	Modern contraceptive prevalence rate (SP.DYN.CONU.ZS); 2007–2024 mean (World Bank)	Economic (9 miss.)
post_socialist	Indicator: 1 if formerly socialist Eastern European state	Regional

N = 21 countries. V-Dem indices are averaged across 2007–2024. ESS composites are pooled across all available rounds for each country (range: 1 round for Romania to 11 rounds for Belgium, Finland, Hungary, Norway, Poland, Portugal, Slovenia, Spain, Sweden). Missing values are handled natively in XGBoost (Stage 1) and via list-wise deletion in Bayesian models (Stage 2, N = 18).

3.4 Construction of Dataset 3: BCS70 Individual-Level Data

The 1970 British Cohort Study (BCS70) follows approximately 17,000 individuals born in a single week in April 1970 in Britain. Seven sweep series are accessed via the UK Data Service under standard academic end-user licence: birth (SN 2666), age 10 (SN 3723), age 16 (SN 3535, with supplementary arithmetic data SN 6095), age 30 (SN 5558), age 42 (SN 7473), and age 51 (SN 9347). The age 51 sweep was released in 2023. BCS70 standard missing-value codes (−1, −7, −8, −9) are converted to NA across all numeric variables, and duplicate observations within each sweep are removed.

3.4.1 Variable selection and analytical sample

The selection of variables follows the theoretical structure of H4, in which the total effect of education on fertility is decomposed into direct and indirect pathways through attitudinal, economic, and partnership mediators. Pre-treatment confounders are drawn from sweeps prior to the treatment (NVQ qualifications measured at age 30); mediators from sweeps at or after the treatment; and the outcome from the age 42 fertility sweep. Table 3.5 sets out all variables.

Table 3.5. Variables in Dataset 3 (BCS70 analytical sample)

Role	Variable	Sweep	Definition
Outcome	*y*	Age 42	Total children born (bd9totce); fertility is largely but not fully complete at age 42 for the 1970 cohort
Treatmen t	*d*	Age 30	Highest NVQ qualification (hinvq00), levels 0–4; level 5 collapsed into level 4 to ensure adequate cell sizes
Confound er	*soc_birth*	Birth	Father's social class at birth (reverse-coded; higher = higher SES)
Confound er	*mage, mage_fb*	Birth	Mother's age at this birth and at first birth
Confound er	*parity, birth_order*	Birth	Number of previous live births; cohort member's birth order

Confounder	*cog_10*	Age 10	Standardised composite cognitive score (reading, vocabulary, arithmetic)
Confounder	*soc_10, inc_10*	Age 10	Household social class and family income at age 10
Confounder	*soc_16, malaise_16*	Age 16	Household social class and Rutter Malaise inventory score at age 16
Mediator (att.)	*m_trad_gender*	Age 30	Mean of three traditional gender-role items (wm1–wm3); contemporaneous with treatment
Mediator (att.)	*m_profamily*	Age 30	Mean of two pro-family attitude items (c1, c3); contemporaneous with treatment
Mediator (econ.)	*m_log_earn*	Age 42	Natural log of gross annual earnings (b9groa); contemporaneous with outcome
Mediator (partner)	*m_partner*	Age 42	Indicator for current cohabiting partner (bd9partp); contemporaneous with outcome

NVQ = National Vocational Qualification. All standard BCS70 missing codes (−1, −7, −8, −9) converted to NA. The treatment collapses NVQ level 5 into level 4. Attitudinal mediators (m_trad_gender, m_profamily) are measured at age 30, contemporaneous with the treatment rather than after it; this timing limitation is discussed in Section 3.8.2. Economic and partnership mediators (m_log_earn, m_partner) are measured at age 42, contemporaneous with the outcome; they post-date the treatment but precede or coincide with the outcome.

The analytical sample is restricted to cohort members with non-missing values on the outcome (y), treatment (d), and sex. This yields 8,318 individuals: 4,445 women and 3,873 men. The treatment distribution within the analytical sample is skewed toward higher qualifications, reflecting the substantial expansion of post-secondary education across the 1970 cohort's lifetime: 11.0% have no qualifications (NVQ 0), 8.1% NVQ 1, 30.9% NVQ 2, 14.4% NVQ 3, and 35.6% NVQ 4. The outcome distribution shows that 21.7% of cohort members are childless at age 42 (y = 0), consistent with published estimates for this cohort (Berrington, 2017).

Pre-treatment confounders with at least 30% non-missing coverage in the analytical sample are retained. Most confounders substantially exceed this threshold: birth characteristics (soc_birth, mage, mage_fb, parity, birth_order) achieve 92–93% coverage; age 10 measures (cog_10, soc_10, inc_10) achieve 64–86% coverage; and age 16 measures (soc_16, malaise_16) achieve 53–77% coverage. The adolescent cognitive score at age 16 (cog_16) achieves only 22% coverage owing to loss to follow-up at the age 16 sweep and is therefore excluded from the H4 confounder set. The final confounder vector for the total-effect models comprises ten variables: soc_birth, mage, mage_fb, parity, birth_order, cog_10, soc_10, inc_10, soc_16, malaise_16.

Mediator coverage is high: attitudinal mediators (m_trad_gender, m_profamily) achieve 99% coverage; the economic mediator (m_log_earn) achieves 70%; and the partnership mediator (m_partner) achieves 78%. Because the mediation models add mediators sequentially to the confounder set (Section 3.8), each stage uses a complete-case sample on the covariates present at that stage. Sample sizes decline progressively: from N = 1,461 women at M0 to N = 1,453 at M1 (adding attitudinal mediators), N = 1,046 at M2 (adding earnings), and N = 812 at M3 (adding partner status). This attrition is discussed as a limitation in Chapter 7.

3.5 Methods for H1: Two-Way Fixed-Effects Panel Regression

H1 holds that the education–fertility relationship is negative across contemporary European populations. The hypothesis is tested with a two-way fixed-effects panel regression of log eTFR on education category, with country and year fixed effects:

$$\log(\mathrm{eTFR}_{r\gamma r}) = \alpha_r + \gamma_\gamma + \beta\cdot\mathrm{education}_{r\gamma r} + \varepsilon_{r\gamma r}$$

where *c* indexes country, *y* year, and *t* education tier; α_r and γ_γ are country and year fixed effects; and $\varepsilon_{r\gamma r}$ is the idiosyncratic error. Standard errors are clustered at the country level. The model is estimated using the fixest R package.

Two specification choices require comment. First, the outcome is log eTFR rather than eTFR itself. The eTFR is bounded below at zero and right-skewed across education groups; the log linearises the multiplicative structure of the standard fixed-effects model and produces coefficients interpretable as percentage differences in fertility across education categories. Second, the reference category is medium rather than low. This choice is motivated by the finding in Chapter 4 that medium-educated women have the lowest fertility in twelve of the twenty-one countries, making medium the most stable and analytically informative baseline. With medium as the reference, the two model coefficients (low vs medium and high vs medium) directly quantify the U-shape. The standard H1 contrast (high vs low) is recovered as a Wald test or by re-estimating with low as reference, yielding $\beta = -0.187$ ($p = 0.002$).

Cluster-robust inference with only 21 clusters relies on asymptotic approximations that may be imprecise at small cluster counts. The wild-cluster bootstrap is recommended for this setting (Cameron, Gelbach, and Miller, 2008) but was not run in the present analysis because the fwildclusterboot package was not available in the analysis environment. The cluster-robust standard errors are therefore the primary inferential tool for H1, with the caveat that p-values should be interpreted with some caution.

3.6 Methods for H2: Quadratic Specification and Quantile Regression

H2 holds that the education–fertility relationship is non-monotonic. Two complementary specifications test this. The first replaces the categorical education factor with an ordinal numeric coding (low = 0, medium = 1, high = 2) and adds a quadratic term:

$$\log(\mathrm{eTFR}_{r\gamma e}) = \alpha_r + \gamma_\gamma + \beta_1 \cdot \mathrm{educ}_{r\gamma e} + \beta_2 \cdot \mathrm{educ}_{r\gamma e}^2 + \varepsilon_{r\gamma e}$$

A positive and statistically significant β_2 indicates U-shaped curvature. The minimum of the curve occurs at education = $-\beta_1 / (2\beta_2)$. The Wald test for $\beta_2 = 0$ is reported alongside an AIC comparison against the linear specification. The quadratic coding assumes equal spacing between the three education tiers (the low–medium and medium–high gaps are both one unit); this is a simplification that should be borne in mind when interpreting the location of the curve minimum.

The second specification uses quantile regression, estimated at $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$, using the quantreg R package with country and year fixed effects entered as dummy variables. The quantile models use high as the reference category for education (the default alphabetical ordering in quantreg), unlike the fixed-effects models in Section 3.5 which use medium as reference; this difference in reference coding is noted where results are reported in Chapter 5. Differences in education coefficients across quantiles reveal whether the non-monotonic gradient is stable across the distribution of fertility or concentrated at particular fertility levels.

3.7 Methods for H3: Two-Stage Discovery–Inference Pipeline

H3 holds that cross-country variation in gradient steepness is better predicted by cultural than by economic indicators. The country sample of N = 21 presents a severe degrees-of-freedom constraint: a traditional regression with twelve covariates would exhaust the available degrees of freedom and be prone to overfitting. This dissertation therefore deploys a two-stage discovery–inference pipeline. Stage 1 uses gradient-boosted regression trees with SHAP value interpretation to identify which features carry predictive signal in a flexible non-parametric framework. Stage 2 uses Bayesian linear regression with regularising priors and proper uncertainty quantification on a parsimonious set of models. The two stages are conceptually distinct: Stage 1 is exploratory and oriented toward the relative importance of feature groups; Stage 2 is confirmatory and oriented toward inferential conclusions about specific coefficients.

3.7.1 Stage 1: XGBoost with SHAP interpretation

Gradient-boosted regression trees are fitted to the gradient_steepness outcome using XGBoost on all 12 predictors (3 ESS attitudinal, 5 V-Dem institutional, 3 economic, 1 regional). XGBoost handles missing values natively, so all 21 countries enter the training data. Hyperparameters are deliberately conservative given the small sample: maximum tree depth is 2, learning rate 0.1, L2 regularisation $\lambda = 5$, L1 regularisation $\alpha = 1$, and minimum child weight 3. The optimal number of boosting rounds is selected by 5-fold cross-validation with early stopping at 20 rounds without improvement.

Out-of-sample predictive performance is assessed via leave-one-out cross-validation (LOOCV), in which the model is refit 21 times with each country omitted in turn. The LOOCV R^2 measures honest predictive power and provides a sanity check on whether macro-level indicators carry generalisable signal. Feature importance is assessed via SHAP values (Lundberg and Lee, 2017) computed using XGBoost's native predcontrib mode. SHAP values decompose each prediction into per-feature contributions, supporting a three-way aggregate comparison: attitudinal-cultural (ESS) versus institutional (V-Dem) versus economic (OECD/World Bank) shares of total mean absolute SHAP.

3.7.2 Stage 2: Bayesian linear models with regularising priors

Six Bayesian linear models are fitted using the brms R package on the complete-case subset of N = 18 countries (dropping North Macedonia, Serbia, and Türkiye, which have missing GDP). All continuous predictors are standardised to z-scores within the complete-case sample so that coefficients are directly comparable across models and features. The six models are: (1) Institutional, with gender_equal and relig_free; (2) Economic, with gdp_percap and flfp; (3) Post-socialist only; (4) Original combined, with gender_equal, flfp, and post_socialist; (5) Attitudinal, with ess_secular and ess_gender_egal; and (6) Updated combined, with ess_secular, gdp_percap, and post_socialist. The attitudinal model (M5) and updated combined model (M6) were added as part of the ESS integration described in Section 3.3.2.

Regularising Student-t(3, 0, 1) priors are placed on all coefficients, the intercept, and the residual standard deviation. The t(3) prior allows for moderately large effects while penalising implausibly extreme coefficients, which is essential at N = 18 to prevent the posterior from being unduly influenced by leverage points. Four chains of 4,000 iterations each, with 2,000 warm-up draws, are run using Stan's No-U-Turn sampler. All six models converged satisfactorily: maximum R-hat across all models and parameters was 1.003, and the minimum effective-sample-size ratio was 0.377. Models are compared on the Watanabe–Akaike Information Criterion (WAIC) and approximate leave-one-out cross-validation (PSIS-LOO).

3.8 Methods for H4: Double Machine Learning Causal Mediation

H4 holds that the effect of education on fertility is mediated primarily by attitudinal rather than economic pathways. Testing this at the individual level faces two methodological challenges. First, education is endogenous: unobserved characteristics such as cognitive ability and family background may drive both educational attainment and fertility. Second, mediation decomposition requires controlling for all confounders of the mediator–outcome relationship, which is high-dimensional in the BCS70 setting.

3.8.1 Partially linear regression via double machine learning

Both challenges are addressed using the Double Machine Learning (DML) framework (Chernozhukov et al., 2018), implemented in the DoubleML R package. The total effect of education on fertility is estimated under a partially linear regression model:

$$Y = \theta \cdot D + g(X) + \varepsilon, \qquad E[\varepsilon \mid X, D] = 0$$

where Y is completed fertility, D is NVQ level (0–4), X is the vector of ten pre-treatment confounders, and θ is the average causal effect of one additional NVQ level on completed fertility. The nuisance functions $g(X) = E[Y|X]$ and $m(X) = E[D|X]$ are estimated using random forests (ranger, 500 trees, minimum node size 5) on a separate data fold. The treatment and outcome are residualised against their conditional expectations, and θ is estimated by OLS regression of the residualised outcome on the residualised treatment (the partialling-out moment condition). Five-fold cross-fitting with three repetitions is used; the final estimate is the median across repetitions. This yields a $\sqrt{N}$-consistent and asymptotically normal estimator of θ that is robust to misspecification of either nuisance function provided the other is consistently estimated.

3.8.2 Sequential mediation decomposition

The total effect is decomposed into pathway-specific components by running the DML procedure four times, sequentially adding mediator groups to the confounder set:

M0 (total effect): $Y = \theta^{T} \cdot D + \mathrm{g}(X) + \varepsilon$

M1 (+attitudes): $Y = \theta_1 \cdot D + \mathrm{g}(X, \mathrm{M}^{\mathrm{aTT}}) + \varepsilon$

M2 (+earnings): $Y = \theta_2 \cdot D + \mathrm{g}(X, \mathrm{M}^{\mathrm{aTT}}, \mathrm{M}^{\mathrm{eNO}}) + \varepsilon$

M3 (+partner): $Y = \theta_3 \cdot D + \mathrm{g}(X, \mathrm{M}^{\mathrm{aTT}}, \mathrm{M}^{\mathrm{eNO}}, \mathrm{M}^{\mathrm{ONTT}}) + \varepsilon$

The indirect effect through each pathway is recovered by subtraction: the attitudinal indirect effect is $\theta^{\mathrm{T}} - \theta_1$, the earnings indirect effect is $\theta_1 - \theta_2$, and the partnership indirect effect is $\theta_2 - \theta_3$. The residual θ_3 represents the controlled direct effect. This sequential approach approximates the natural indirect effect under sequential ignorability (Imai, Keele, and Tingley, 2010), with the standard caveat that the ordering of mediator blocks affects the decomposition. Attitudes are entered first because they are measured at age 30 (the same sweep as the treatment) and are most plausibly causally prior to labour-market outcomes in the life course, even though their contemporaneous measurement with the treatment is a limitation.

Two additional timing caveats are acknowledged. The attitudinal mediators (m_trad_gender, m_profamily) are measured at age 30, the same sweep from which NVQ qualifications are drawn; they are therefore contemporaneous with the treatment rather than temporally downstream of it. The sequential ignorability assumption requires that mediators are not themselves affected by unmeasured common causes of the mediator and outcome, which is stronger when mediator and treatment are contemporaneous. The economic mediator (m_log_earn) and partnership mediator (m_partner) are measured at age 42, the same sweep as the outcome; they post-date the treatment but are contemporaneous with the outcome, meaning the DML model controls for them as additional confounders of the education–fertility relationship rather than as true downstream mediators in the strict causal sense.

Mediation models are estimated separately for women and men, as fertility patterns and labour-market dynamics differ substantially by sex (Berrington, 2017). A pooled model with sex as an additional confounder is reported for the total-effect estimate as a consistency check.

3.9 Robustness Checks

Robustness checks are organised by hypothesis. For H1, four checks are reported. First, a balanced subsample restricted to the eight countries with the full 2007–2024 panel (Croatia, Czechia, Finland, Greece, Hungary, Portugal, Romania, Slovakia). Second, exclusion of pandemic years 2020 and 2021, which produced atypical fertility patterns across most European countries. Third, exclusion of the eight composition-driven U-shape countries (Croatia, Czechia, Estonia, Finland, Latvia, Poland, Slovakia, Slovenia), to assess whether the H1 result is driven by their potentially compositionally inflated low tiers. Fourth, re-estimation with low rather than medium as the reference category, as a sensitivity check on the coefficient framing. A fifth robustness check weights each country–year–education observation by the implied female population in that cell, constructed by summing the education-specific population denominators from the ASFR data across age groups. The population-weighted model yields a low versus high coefficient of +0.242 ($p = 0.006$), confirming that the negative low–high gradient is robust to differential country size. However, the medium versus high coefficient under weighting is −0.012 ($p = 0.890$), compared to −0.102 ($p = 0.022$) in the unweighted model. The U-shape finding — specifically the lower fertility of medium-educated relative to high-educated women — is therefore sensitive to population weighting and is not robust in larger countries. This qualification is discussed alongside the main H1 results in Chapter 5.

A wild-cluster bootstrap was planned to supplement the cluster-robust standard errors given the small number of clusters ($N = 21$), following Cameron, Gelbach, and Miller (2008). This could not be implemented because the fwildclusterboot package is not currently available for R 4.5.1. The cluster-robust standard errors reported throughout should therefore be interpreted with the caveat that asymptotic approximations may be imprecise at small cluster counts, though the primary coefficients are sufficiently large relative to their standard errors that this concern is unlikely to alter the substantive conclusions.

For H2, the parametric quadratic specification is complemented by quantile regression at five quantiles ($\tau = 0.10, 0.25, 0.50, 0.75, 0.90$), providing a non-parametric distributional check on the U-shape across the conditional distribution of log eTFR.

For H3, the central robustness observation is the disagreement between the Stage 1 and Stage 2 results. Stage 1 (XGBoost SHAP) assigns 59.5% of aggregate importance to cultural indicators combined (54.1% institutional, 5.4% attitudinal) and 40.5% to economic indicators. Stage 2 (Bayesian model comparison) ranks the economic model best on both WAIC and LOO-IC, with GDP per capita as the strongest individual predictor (posterior P(negative) = 96.6%). Rather than suppressing this disagreement, it is treated as a substantive finding: the LOOCV R^2 of −0.086 indicates that macro indicators collectively lack reliable out-of-sample predictive power at $N = 21$, and the model comparison uncertainty (all WAIC differences within two standard errors) reflects genuine inferential limitation at this sample size.

For H4, three robustness checks are deployed. First, the analysis is run separately for women and men, and the pooled estimate is compared to the sex-stratified estimates. Second, the complete-case sample for the mediation stages is acknowledged as an attrition concern: N drops from 1,461 (M0) to 812 (M3) for women as mediators with incomplete coverage are added sequentially. Third, the mediator ordering is noted as a sensitivity: reversing the block order (earnings first, then attitudes, then partnership) changes the decomposition mechanically but does not alter the qualitative conclusion that the attitudinal pathway accounts for a small share of the total effect.

3.10 Software, Reproducibility, and Ethical Considerations

All analyses are conducted in R version 4.5.1. Primary packages: tidyverse and data.table for data manipulation; eurostat for Eurostat API access; fixest (Bergé, 2018) for panel fixed-effects regression; quantreg (Koenker, 2005) for quantile regression; xgboost (Chen and Guestrin, 2016) for gradient boosting; brms (Bürkner, 2017) and Stan for Bayesian regression; loo (Vehtari, Gelman, and Gabry, 2017) for model comparison; DoubleML (Bach et al., 2021) for double machine learning; ranger (Wright and Ziegler, 2017) for random forest nuisance estimation; countrycode for country-name harmonisation; and ggplot2 for visualisation. The project is organised as a numbered-script pipeline (00–08) in which each script takes named inputs from data/derived or data/raw and writes named outputs back, supporting reproducibility.

All datasets are either fully open-access or available under standard academic-research licence. Eurostat, V-Dem, OECD, and World Bank data are downloaded from their respective open data portals. ESS data are read from a locally stored integrated file obtained via the ESS data portal under academic registration. BCS70 data are accessed via the UK Data Service under an end-user licence filed for this dissertation project; all data handling complies with UKDS access conditions. No new data are collected from human participants, and no identifiable individuals are analysed or reported. Ethical approval was obtained through Trinity College Dublin's standard process for secondary data analysis.

Two transparency notes are warranted. First, this chapter documents three departures from the original research proposal openly: the shift from Wittgenstein cohort fertility to Eurostat period eTFR, the reduction in country sample from 35 to 21, and the addition of ESS attitudinal predictors as a methodological refinement during analysis. Second, the empirical findings are interpreted with appropriate epistemic humility given the small effective country sample for H3 (N = 21 for XGBoost, N = 18 for Bayesian) and the cohort-specificity of H4 (a single British 1970 birth cohort). These limitations are addressed substantively in Chapter 7.